

CONTRIBUTING TO FINOMOP MAPPINGS

A clinician friendly guide

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Last updated December 2024

- Changes and new mappings are done using USAGI.
- This guide is designed to be as accessible as possible, not requiring previous experience of git or version control, Usagi, and so on. However, basic knowledge of OMOP is required. Things like what is a vocabulary will not be discussed here.

THIS GUIDE IS ARRANGED AS FOLLOWS:

- 1) Set up Usagi and the required files.
- 2) Do changes to mappings
- 3) Use github / version control

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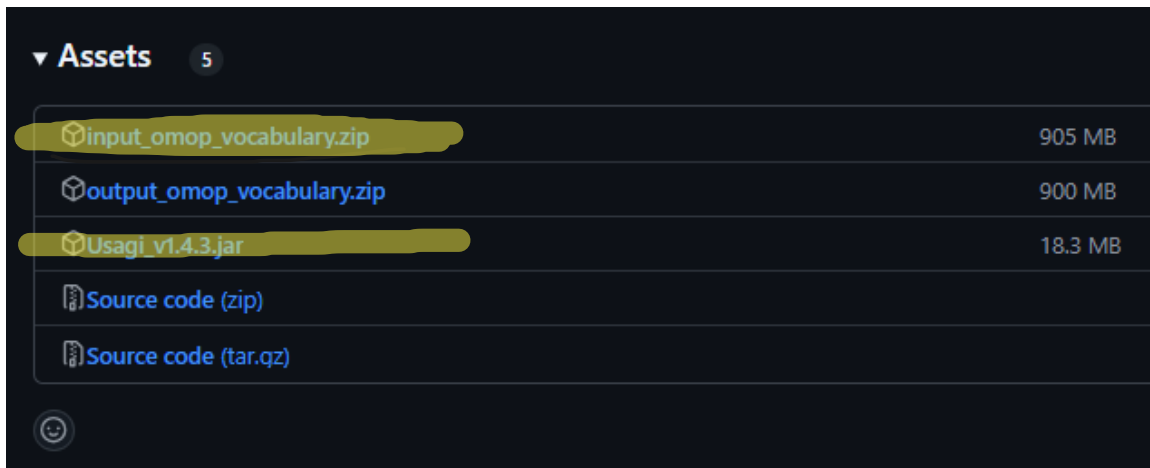
More detailed documentation exists in

https://github.com/FinOMOP/FinOMOP_OMOP_vocabulary/blob/development/DOCUMENTATION/

the FINOMOP Github and this slideshow is just another,
more visual guide to contributing.

USAGI SETUP

- You need: **1) OHDSI vocabularies**, **2) USAGI** and **Java** (see end of slide)
- These are located in the below repository under /releases/
- input_omop_vocabulary.zip
- Download the **1) input_omop_vocabulary.zip** and **2) Usagi_vX.X.X.jar** and **input_omop_vocabulary.zip** files from https://github.com/FinOMOP/FinOMOP_OMOP_vocabulary/releases/ (visited 12.12.2024, VN)
- For detailed instructions on using and installing USAGI see end of slide



A pic highlighting the assets you need to download from FINOMOP repo.

- Get Java if you dont have it, for ex., from <https://openjdk.org/> following the directions there. FINOMOP has nothing to do with Java.
- Usagi can also be downloaded from OHDSI. FINOMOP did not develop USAGI, but OHDSI did and its a part of the OHDSI standard ETL toolstack <https://ohdsi.github.io/Usagi/installation.html>.

USAGI SETUP 2:

Launching Usagi and loading vocabularies

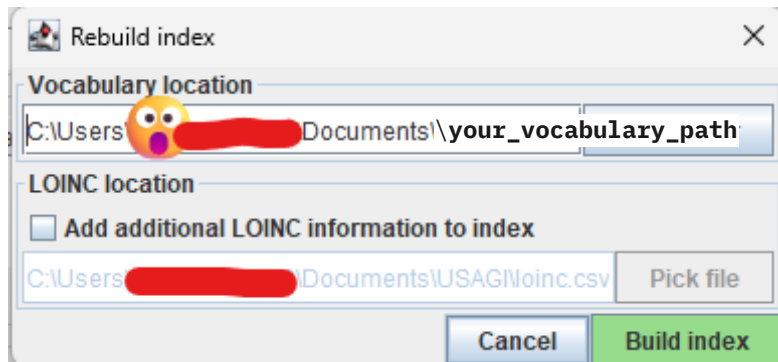
Launching Usagi

- Usagi is a .jar file. You open it with Java.
- For example on Windows, if you downloaded and installed OpenJDK you right-click on the Usagi file, choose Open with, and point to where java.exe is. Typically it is in a folder under Program Files like *C:\Program Files\Java\jdk-23.0.1\bin*. For Linux or further support see for ex., <https://askubuntu.com/questions/322458/how-to-open-jar-file-using-open-jdk>
- The author starts Usagi from windows commandline by calling `C:\Users\XXXX\Documents\USAGI> java -jar .\Usagi_v1.4.3.jar`

USAGI SETUP 2:

Loading vocabularies

- Once you open Usagi, it will kindly ask you for the location of the vocabularies you want to load.
- Point this to where you saved and unzipped the vocabularies. (input_omop_vocabulary file)



Building index. This will take a while....
Loading parent-child information

Wait for the index to build.

After you've loaded the vocabularies you can use USAGI to make your own, new or proposed mappings.

If this was your goal, then you can stop reading this tutorial here!

If you are adding a completely new vocabulary see below link.

https://github.com/FinOMOP/FinOMOP_OMOP_vocabulary/blob/development/DOCUMENTATION/workflow_add_a_new_vocabulary.md

USE CASE OF USAGI: CONTRIBUTING TO FINOMOP MAPPING / VOCABULARIES

First of all, if you've come this far, you should be attending the vocabulary workgroup monthly meetings.

Ask xxxx for an invite.

Here we depict the process through 2 real-world example cases:

- Updating ICD10fi C codes that are no longer standard
- Adding new mappings for ICD10fi codes

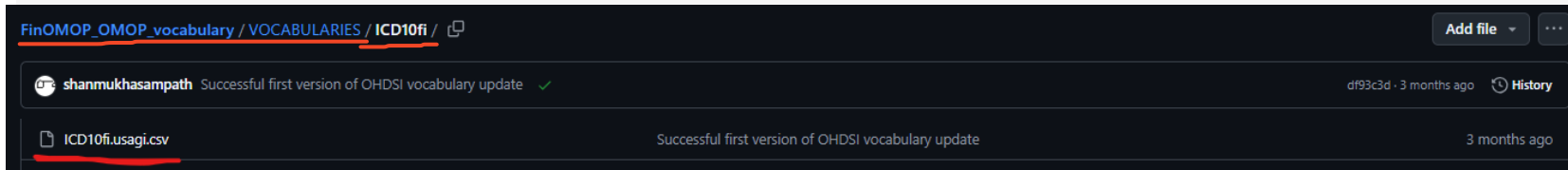
The whole process has roughly the following steps, (not taking into account finding the codes themselves which is presumed to be done)

- 1) The updating and or- modification or creation of USAGI files.
- 2) Version control: creating an issue and a branch in github.
https://github.com/FinOMOP/FinOMOP_OMOP_vocabulary/blob/development/DOCUMENTATION/repo_rules_and_nomenclature.md
- 3) Testing and building the vocabulary (not depicted in this slideshow)

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

Here our issue that some codes we have previously mapped to from ICDfi C (cancer) codes **are no longer standard** after an OHDSI vocabulary update that introduced the cancer modifier vocab.

1) Get the vocabulary you want to modify.



2) Open the file in usagi with File -> Open

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

Here are a few of the relevant columns for the codes we want to update.

sourceCode	sourceName	domain	conceptId	conceptName
C78.30	Secondary malignant tumor of the nose or its sinuses	Condition	4312557	Secondary malignant neoplasm of nose
C78.31	Secondary malignant tumor of the middle ear or inner ear	Condition	4312549	Secondary malignant neoplasm of middle ear
C78.39	Secondary malignant neoplasm of other and unspecified respiratory organs	Condition	253717	Secondary malignant neoplasm of respiratory tract
C79.30	Secondary malignant tumor of the cerebrum	Condition	4311763	Secondary malignant neoplasm of cerebrum
C79.31	Secondary malignant tumor of the cerebellum	Condition	4312289	Secondary malignant neoplasm of cerebellum
C79.32	Secondary malignant tumor of the brainstem	Condition	4312280	Secondary malignant neoplasm of brain stem
C79.39	Secondary malignant tumor of the meninges, ventricle or other part of the brain	Condition	4312290	Secondary malignant neoplasm of cerebral meninges
C79.40	Secondary malignant tumor of the spinal cord	Condition	4311342	Secondary malignant neoplasm of spinal cord
C79.41	Secondary malignant tumor of optic nerve and eye	Condition	4312954	Secondary malignant neoplasm of eye
C79.42	Secondary malignant tumor of the acoustic nerve	Condition	4312039	Secondary malignant neoplasm of acoustic nerve
C79.49	Secondary malignant neoplasm of other and unspecified parts of nervous system	Condition	373425	Secondary malignant neoplasm of nervous system
C79.5+M49.5	Metastatic fracture of vertebra	Condition	3655619	Pathologic fracture of vertebra at site of metastatic neoplasm

Lets pick one as an example few that we will show as an example.

sourceCode	sourceName	domain	conceptId	conceptName
C78.30	Secondary malignant tumor of the nose or its sinuses	Condition	4312557	Secondary malignant neoplasm of nose

The concept has been made non-standard.

Concept ID	4312557	athena.ohds.org
Concept code	94445004	Athena
Validity	Valid	
Concept	Non-standard	

Non-standard to Standard map (OMOP) [Metastasis to nose](#) 35226213 Cancer Modifier

conceptId	conceptName	domainId
35226213	Metastasis to nose	Measurement

We want to change the concept and domain.

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

Usagi - ICD10fi.usagi.csv

Status	Source code	Source term
Approved	C78.30	Secondary malignant...

First, it can be good for clarity to put the status of concepts you're about to modify to "unapproved" from Edit > Unapprove

Usagi - ICD10fi.usagi.csv

File	Edit	View	Help
	Unapprove		Alt-A
	Clear selected		
	Assign Reviewers		

Unchecked

C78.30

Secondary malignant...

2000501202

This will help you keep track of what you have done so far.

Lets pick one as an example few that we will show as an example.

sourceCode	sourceName	domain	conceptId	conceptName
C78.30	Secondary malignant tumor of the nose or its sinuses	Condition	4312557	Secondary malignant neoplasm of nose

The concept has been made non-standard.

Concept ID: 4312557
Concept code: 94443004
Validity: Valid
Concept: Non-standard

Non-standard to Standard map (CMOP) Metastasis to nose 35226213 Cancer Modifier

conceptId	conceptName	domainId
35226213	Metastasis to nose	Measurement

We want to change the concept and domain.

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

Source code

Source code	Source term	Frequency	sourceConceptId	sourceName_fi	sourceConceptClass	sourceDomain	sourceValidStartDate	sourceValidEndDate	sourceParents	sourceParentVocabulary	obsolete_conceptId
C78.30	Secondary malignant tu... -1		2000501202	Etapäsäke nenässä tai ...	ICD10fi 1 Code	Condition	1900-01-01	2099-12-31	C78	ICD10	

Target concepts

Concept ID	Concept name	Domain	Concept class	Vocabulary	Concept code	Standard concept	Parents	Children	Mapping Type	Creation Provenance
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1) You can then use Usagi to find a concept.

2) If you already know a bit about what youre looking for its useful to filter.

Search

Query

☐ Use source term

☒ Query: Metastasis nose

Filters

☐ Filter by user selected concepts / ATC code

☒ Filter standard concepts

☒ Include source terms

☐ Filter by concept class:

☒ Filter by vocabulary: Cancer Modifier

☐ Filter by domain:

Results

Score	Term	Concept ID	Concept name	Domain	Concept class	Vocabulary	Concept code	Standard concept	Parents	Children
0.75	Metastasis to nose	35226213	Metastasis to nose	Measurement	Metastasis	Cancer Modifier	OMOP5031760	S	1	1
0.73	Metastasis	36769180	Metastasis	Measurement	Metastasis	Cancer Modifier	OMOP4998856	S	0	42
0.62	Metastasis to skin of no...	35225684	Metastasis to skin of nose	Measurement	Metastasis	Cancer Modifier	OMOP5031876	S	1	0
0.58	Metastasis to septum o...	35225668	Metastasis to septum of nose	Measurement	Metastasis	Cancer Modifier	OMOP5031860	S	1	0
0.57	Metastasis to cartilage ...	35226095	Metastasis to cartilage of nose	Measurement	Metastasis	Cancer Modifier	OMOP5031561	S	1	0
0.57	Metastasis to arm	35226032	Metastasis to arm	Measurement	Metastasis	Cancer Modifier	OMOP5031496	S	1	1
0.57	Metastasis to vestibule ...	35225595	Metastasis to vestibule of nose	Measurement	Metastasis	Cancer Modifier	OMOP5032004	S	1	0
0.56	Metastasis to bone	36769301	Metastasis to bone	Measurement	Metastasis	Cancer Modifier	OMOP4998978	S	1	9
0.54	Metastasis to lip	35225636	Metastasis to lip	Measurement	Metastasis	Cancer Modifier	OMOP5031700	S	1	3
0.54	Metastasis to carina	35226092	Metastasis to carina	Measurement	Metastasis	Cancer Modifier	OMOP5031558	S	1	0
0.54	Metastasis to anus	35226026	Metastasis to anus	Measurement	Metastasis	Cancer Modifier	OMOP5031490	S	1	0
0.54	Metastasis to hand	35226163	Metastasis to hand	Measurement	Metastasis	Cancer Modifier	OMOP5031642	S	1	1
0.54	Metastasis to penis	35226247	Metastasis to penis	Measurement	Metastasis	Cancer Modifier	OMOP5031708	S	1	2

3) When you have found the concept you think is right click

Replace concept Add concept

Comment: Changed from SNOMED code that was de-standardized when Cancer Modifier was introduced, replacing Pko 2022 mapping.

Its good to comment on what youve done!

CONTRIBUTING TO FINOMOP VOCABULARIES: UPDATING AN EXISTING MAPPING

Replace concept

Add concept

Flag

UNREVIEWED

Approve

Once youve made changes, you can flag the row.

Author: VN Vocabulary version: v5.0 30-AUG-24

Approved	Medullary thyroid carcinoma (thyroid)	2000501200	Kipiraashasan med.	ICD10R 1 Code	Condition	100-01-01	2000-12-31	C73	ICD10	0.00	4111011	Medullary thyroid c.	Condition	Disorder	SNOMED	255533005	8	2	1	EQUAL	PNv (2022-08-26)	C73.94
Approved	Malignant tumor of the thyroid gland (thyroid gland), other or unspecified histology	2000501201	Kipiraashasan med.	ICD10R 1 Code	Condition	100-01-01	2000-12-31	C73	ICD10	0.00	125004	Primary malignant	Condition	Disorder	SNOMED	9400005	8	3	153	NEVER	PNv (2022-08-26)	C73.98
PLUGGED	Secondary malignant tumor of the nose or its sinuses	2000501202	Elipsesake rekusu.	ICD10R 1 Code	Condition	100-01-01	2000-12-31	C78	ICD10	0.00	2525213	Metastasis to nose	Measurement	Metastasis	Cancer Modifier	0840P2031150	8	1	1	Changed from SNOMED	VN (2024-12-12)	C78.32
Unchecked	Secondary malignant tumor of the middle ear or inner ear	2000501203	Elipsesake rekusu.	ICD10R 1 Code	Condition	100-01-01	2000-12-31	C78	ICD10	0.14	4121548	Metastatic maligna.	Condition	Disorder	SNOMED	94413009	0	0	0			C78.31
Unchecked	Secondary malignant neoplasm of other and unspecified respiratory organs	2000501204	Elipsesake rekusu.	ICD10R 1 Code	Condition	100-01-01	2000-12-31	C78	ICD10	0.00	253717	Metastatic maligna.	Condition	Disorder	SNOMED	94010004	0	0	0			C78.29

Now lets pretend we would be ready to submit to FINOMOP.

TODO:

ISSUE.....github and so on